Vince Kalinowski

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Education

University of Texas Rio Grande Valley — Edinburg, TX
Bachelor of Science in Computer Science — *Sep. 2024 – Dec. 2026*

Grand Rapids Community College — Grand Rapids, MI
Aug. 2022 – May 2024

Experience

Undergraduate Research Assistant — May 2026 – Present
Algorithmic Self-Assembly Research Group, University of Texas Rio Grande Valley — *Edinburg, TX*

- Reviewed algorithmic self-assembly literature at a rate of one paper per week, presenting findings and open questions during weekly research meetings.
- Collaborated with faculty and student researchers to evaluate unconventional computational approaches to open self-assembly problems, contributing to the group's ongoing research direction.

Projects

UFC Prediction Model | *Python, Pandas, Scikit-Learn* — Apr. 2026

- Built a UFC fight prediction system using a Random Forest classifier trained on historical UFC fight data from 2012–2025.
- Engineered matchup-based features including age, reach, height, and striking differentials, achieving 66.67% accuracy across a held-out test set of 1,455 fights.

Diagrammed | *React, Django, React Flow, Supabase, Render, OpenAI API* — Jul. 2026

- Developed an interview prep platform for practicing system design interviews through interactive architecture diagrams and AI-generated feedback.
- Built a React Flow interface for designing distributed systems and a Django backend for authentication, practice sessions, and persistent diagram data.

FeynmanAI | *Python, Electron, OpenAI API* — Jan. 2026

- Developed an AI-powered desktop tutoring application implementing the Feynman Technique through interactive explanations and adaptive questioning.
- Designed prompts that generate Socratic follow-up questions targeting gaps in a learner's explanation and synthesize multi-turn conversations into end-of-session feedback.

Technical Skills

Languages: Python, C++, JavaScript, Java, Haskell, HTML, CSS
Frameworks & Libraries: React, Django, Electron, React Flow, NumPy, Pandas, Scikit-Learn
Databases: PostgreSQL, SQLite
Tools & Platforms: Git, Supabase, Vercel, Render